

MSTAR PRACTICE WORKSHEET**Lesson 8-2: Write and Solve Equations Using Addition or Subtraction**

Grade 6 Mathematics · Standard 6.AT.8

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: A**Solve: $x + 9 = 15$

- ☒ **A** $x = 6$
- ☐ **B** $x = 9$
- ☐ **C** $x = 15$
- ☐ **D** $x = 24$

Subtract 9 from both sides: $x = 15 - 9 = 6$. Check: $6 + 9 = 15$ ✓**Question 2** **SELECTED RESPONSE — CORRECT: C**Solve: $y - 7 = 20$

- ☐ **A** $y = 7$
- ☐ **B** $y = 13$
- ☒ **C** $y = 27$
- ☐ **D** $y = 140$

Add 7 to both sides: $y = 20 + 7 = 27$. Check: $27 - 7 = 20$ ✓

Question 3 SELECTED RESPONSE — CORRECT: B

Solve: $n + 15 = 42$

- ☐ A $n = 15$
- ☒ B $n = 27$
- ☐ C $n = 42$
- ☐ D $n = 57$

Subtract 15 from both sides: $n = 42 - 15 = 27$. Check: $27 + 15 = 42$ ✓

Question 4 SELECTED RESPONSE — CORRECT: C

Solve: $m - 11 = 25$

- ☐ A $m = 14$
- ☐ B $m = 25$
- ☒ C $m = 36$
- ☐ D $m = 275$

Add 11 to both sides: $m = 25 + 11 = 36$. Check: $36 - 11 = 25$ ✓

Question 5 SELECTED RESPONSE — CORRECT: D

A shirt costs \$ d . After a \$15 discount, it costs \$38. What is d ?

- ☐ A \$23
- ☐ B \$38
- ☐ C \$45
- ☒ D \$53

$d - 15 = 38 \rightarrow d = 38 + 15 = 53$.

Question 6

SELECTED RESPONSE — CORRECT: B

Unlock Card 1: Solve $x + 7 = 15$.

- ☐ A $x = 7$
- ☒ B $x = 8$
- ☐ C $x = 22$
- ☐ D $x = 105$

Subtract 7 from both sides: $x = 15 - 7 = 8$. Check: $8 + 7 = 15$ ✓

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7

TWO-PART QUESTION

Part A — correct answer: DWhich value of n is the solution of $n - 18 = 27$?

- ☐ A $n = 9$
- ☐ B $n = 18$
- ☐ C $n = 27$
- ☒ D $n = 45$

The variable has 18 subtracted from it, so add 18 to both sides to isolate n : $n = 27 + 18 = 45$. Check by substituting: $45 - 18 = 27$.

- **A:** That subtracts 18 from 27. The variable already had 18 taken away, so ADD it back.
- **B:** That is the amount subtracted, not the solution. Add it to 27.
- **C:** That is the value after 18 was subtracted, not n itself. Add 18 to recover n .

Part B — correct answer: A

Which reasoning justifies the step used to find that answer?

- ☒ A Because 18 is subtracted from n , adding 18 to both sides is the inverse operation that leaves n by itself.
- ☐ B Because the equation shows subtraction, 18 should be subtracted from both sides as well.
- ☐ C Because the number written after the equal sign is always the value of the variable.
- ☐ D Because the smaller number in a subtraction equation is always the solution.

Subtraction is undone by addition. Adding 18 to both sides keeps the equation balanced and isolates the variable, giving $n = 45$. Subtracting 18 from both sides instead would give $27 - 18 = 9$, and $9 - 18$ does not equal 27.

Question 8

WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Adding When the Inverse Is Subtraction

Darius solved the equation $x + 23 = 40$. He wrote: ' $x + 23 = 40$, so $x = 40 + 23$, which means $x = 63$.'

Explain Darius's misconception and give the correct value of x .

Model answer: $x = 17$. The 23 is being added to x , so the inverse operation is subtraction: subtract 23 from both sides to get $x = 40 - 23 = 17$. Check by substituting: $17 + 23 = 40$. Darius's answer fails the same check, because $63 + 23 = 86$, not 40.

2 points	Student explains that adding 23 is undone by subtracting 23 from both sides, not by adding again, and gives $x = 17$.
1 point	Student gives $x = 17$ but does not explain why subtraction is the inverse operation needed.
0 points	Student agrees with Darius or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

A detective solved $x + 25 = 60$ and got $x = 35$. Another detective solved $y - 25 = 60$ and got $y = 35$. Are both correct? Explain using inverse operations and check each answer.

Model answer: Only the first detective is right. For $x + 25 = 60$ the inverse is subtraction, so $x = 60 - 25 = 35$, and $35 + 25 = 60$ checks out. For $y - 25 = 60$ the inverse is addition, so $y = 60 + 25 = 85$, not 35 — substituting 35 gives $35 - 25 = 10$, not 60. Opposite operations need opposite inverses, even when the numbers look identical.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.